Modern Java ile

Design Patterns'ın

Fonksiyonel Evrimi

JavaDay İstanbul 2026

Hakkımda

Ümit Köse

Software Developer

Türksat Uydu Haberleşme Kablo TV ve İşletme A.Ş.

SOSYAL MEDYA

- YouTube: @umiitkose
- X: @umiitkose
- GitHub: @umiitkose

REPO QR

Sunumdaki tum kodlar:
`github.com/umiitkose/javadayistanbul-functional-evolution-of-design-patterns-with-modern-Java`

Gündem

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01 **Java 8 - Java 26**
Fonksiyonel Programlama Yolculuğu

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Someone has already solved your problems.

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Klasik OOP ↔ modern Java (records, lambda, kompozisyon)

Pattern'ler

- 🏗️ Builder → Immutability & Records
- 🎯 Strategy → Lambda & Functional Interfaces
- 🎨 Decorator → Function Composition
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Java 8 · 2014

Lambda + Functional Interfaces

Davranış birinci sınıf; `Consumer`, `Function`, `Predicate`.

FP kapısı açıldı

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Stream API

`filter` → `map` → `reduce`; lazy boru hatları.

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API evrimi

2018-2023 — VERI & TIP SISTEMI

Records · 14→16

Records

Değişmez taşıyıcılar; `equals/hashCode/toString` .

Veri ≠ davranış

Sealed · 17 LTS

Sealed Classes

Kontrollü hiyerarşi; derleyici tüm alt tipleri bilir.

ADT

Java 21 LTS

Switch + record patterns

Eksiksiz dallanma; okunabilir `switch` .

Pattern matching

Java 8 → Java 26

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JAVA 24-26 · FP ILE ÖNE ÇIKANLAR

JEP 485 · Java 24

Stream Gatherers

Stream'e özelleştirilebilir ara `gather` adımları — `map/filter` ötesi FP tarzı boru hatları.

Stream + kompozisyon

JEP 500 · Java 26

Make final mean final

Final alanlar reflection'a karşı gerçekten değişmez.

Design Patterns

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Amaç

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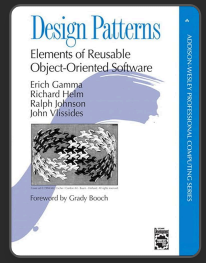

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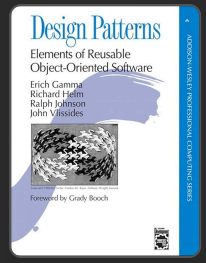

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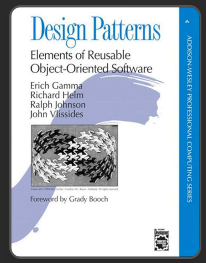

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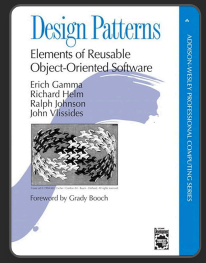

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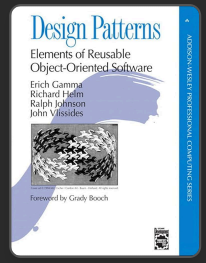

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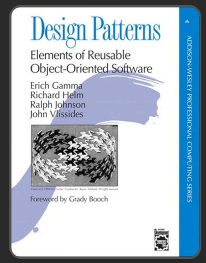

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Modern Java ile Değişim

Pattern'ler aynı kalır; uygulama şekli modernleşir. Lambda, Stream API, records, sealed classes ve pattern matching ile daha az kod, daha net okunabilirlik ve daha kolay test elde edilir.

Sunumdaki Pattern'ler

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Lambda Expressions
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Her pattern için **solda klasik OOP** ve **sağda modern FP** yaklaşımını yan yana göreceğiz

🏗️ Builder Pattern — klasik OOP

Tanım: Karmaşık nesneyi *adım adım* kurup `build()` ile tek seferde doğrulayan creational pattern. Telescoping constructor yerine okunabilir fluent API.

Telescoping

```
new Thing("a","b", null, null, BigDecimal.ZERO);
```

Mutable ara durum

Builder'da fluent setter'lar (`requestedQuantity` vb.) ara alanları değiştirir.

İki katman

Ürün immutable olsa da *kurulum süreci* mutasyon içerir.

Uzun build()

Çok alanda doğrulama tek metotta şişebilir.

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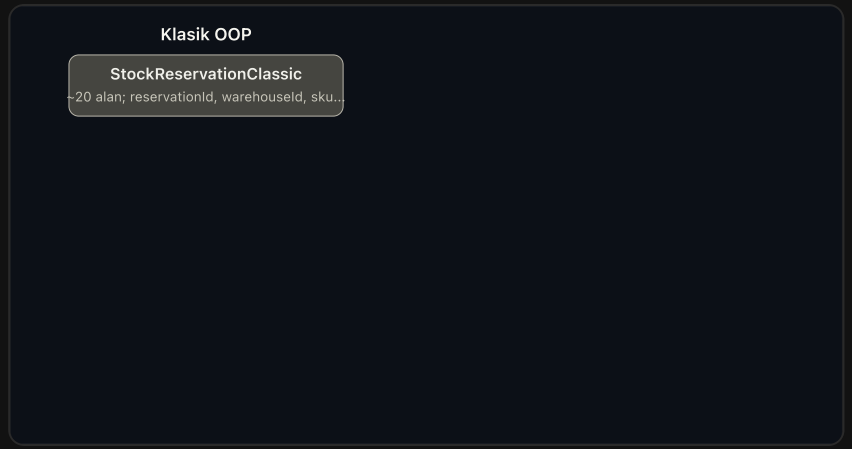

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Modern – record kompozisyonu

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```java
// classic/builder/StockReservationClassic.java — yapı özeti
public final class StockReservationClassic {

    // private final alanlar (~20)

    private StockReservationClassic(Builder builder) {
        // builder alanlarından atama
    }

    // getter'lar …
    @Override public String toString() { /* … */ }
    @Override public boolean equals(Object o) { /* reservationId
    @Override public int hashCode() { return Objects.hash(reserv

    public static class Builder {
        private final String reservationId;
        private final String warehouseId;
        private final String sku;
        private int requestedQuantity;
        // reservedQuantity, unitCost, discountPercent, vatPerce
        // grossWeightKg, netWeightKg, bestBeforeDate, batchNumb

        public Builder(String reservationId, String warehouseId,
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```
// Alt tiplerde validasyon (compact constructor) — örnek: Reserv
public record ReservationIdentity(String reservationId, String w
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// Kullanım: factory / defaults ile birleştirme
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🎯 Strategy Pattern

Tanım: Strategy Pattern, bir algoritma ailesini ayrı sınıflar/fonksiyonlar olarak kapsüller ve çalışma anında uygun olanı seçmeyi sağlar.

🎯 Klasik OOP – Strategy

Modern FP – Strategy

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```java
// classic/strategy/PaymentStrategy.java
public interface PaymentStrategy {
    void pay(BigDecimal amount);
}
```

Modern FP – Strategy

🎯 Klasik OOP – Strategy

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}
```

```java
// StrategyDemo.modernApproach( ... )
Consumer<BigDecimal> cc = PaymentService.creditCard("...0366", "Ahmet Yilmaz");
PaymentService.processPayment(cc, amount);
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// runtime'da farkli strateji secimi
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Strategy davranisini sinif yerine fonksiyonel bir tip (ornegin `Consumer<BigDecimal>`) ile temsil ediyoruz.

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Lambda ve Tip Bagimliligi

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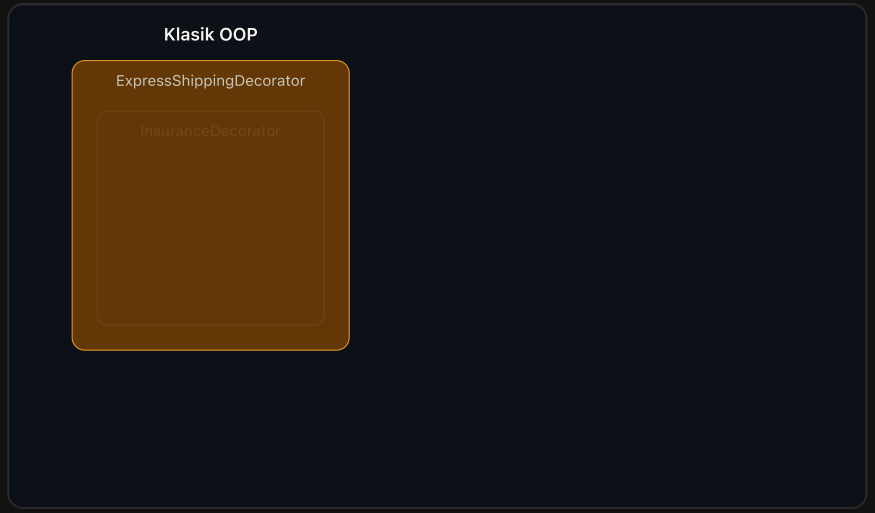
🎨 Decorator Pattern

Tanım: Decorator, bir nesnenin davranışını ana sınıfı değiştirmeden katman katman genişletmeyi sağlayan yapısal bir pattern'dir.



Klasik OOP – Decorator

Modern FP – Decorator

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Modern FP – Decorator

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```java
public class GiftWrapDecorator extends OrderServiceDecorator {
    public GiftWrapDecorator(OrderService wrapped) { super(wrapped); }
    @Override
    public Order process(Order order) {
        Order processed = wrapped.process(order); // once icteki zincir
        return processed.addFeature("Hediye Paketi", new BigDecimal("15"));
    }
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```java
OrderService service = new ExpressShippingDecorator(
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Modern FP – Decorator

```
// modern/decorator/OrderEnhancer.java
public static UnaryOperator<Order> giftWrap() {
    return order -> order.addFeature("Hediye Paketi", new BigDecimal("15"));
}

public static UnaryOperator<Order> insurance() {
    return order -> order.addFeature("Kargo Sigortasi", new BigDecimal("10"));
}

public static UnaryOperator<Order> expressShipping() {
    return order -> order.addFeature("Hizli Kargo", new BigDecimal("25"));
}
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Klasik OOP – Decorator

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}
```

```java
// Normal kullanim (andThen ile acik zincir):
// Function<Order, Order> flow = OrderEnhancer.giftWrap()
//     .andThen(OrderEnhancer.insurance())
//     .andThen(OrderEnhancer.expressShipping());
```

```java
// Sorunlu kullanim:
// UnaryOperator<Order> campaignFlow = campaign.stream()
//     .reduce(UnaryOperator.identity(), UnaryOperator::andThen);
```

```java
// 2) Reduce ile — dinamik, runtime'da liste degisebilir (calisan surum)
List<UnaryOperator<Order>> campaign = List.of(
    OrderEnhancer.giftWrap(),
    OrderEnhancer.insurance(),
    OrderEnhancer.expressShipping()
);
UnaryOperator<Order> enhanceDynamic = campaign.stream()
    .reduce(UnaryOperator.identity(), (pipeline, step) -> pipeline.andThen(step));

var order = new Order("ORD-001", new BigDecimal("200"));
var campaignResult = enhanceDynamic.apply(order);
```

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        Order processed = wrapped.process(order); // once icteki zincir
        return processed.addFeature("Hediye Paketi", new BigDecimal("15"));
    }
}
```

```java
OrderService service = new ExpressShippingDecorator(
    new InsuranceDecorator(
        new GiftWrapDecorator(new BasicOrderService())
    )
);
```

Hap Bilgi 1

Classic yaklaşımda her yeni özellik için yeni bir wrapper sınıfı gerekir.

Modern FP – Decorator

```java
// modern/decorator/OrderEnhancer.java
public static UnaryOperator<Order> giftWrap() {
    return order -> order.addFeature("Hediye Paketi", new BigDecimal("15"));
}

public static UnaryOperator<Order> insurance() {
    return order -> order.addFeature("Kargo Sigortasi", new BigDecimal("10"));
}

public static UnaryOperator<Order> expressShipping() {
    return order -> order.addFeature("Hizli Kargo", new BigDecimal("25"));
}
```

```java
// Normal kullanim (andThen ile acik zincir):
// Function<Order, Order> flow = OrderEnhancer.giftWrap()
//     .andThen(OrderEnhancer.insurance())
//     .andThen(OrderEnhancer.expressShipping());
```

```java
// Sorunlu kullanim:
// UnaryOperator<Order> campaignFlow = campaign.stream()
//     .reduce(UnaryOperator.identity(), UnaryOperator::andThen);
```

```java
// 2) Reduce ile — dinamik, runtime'da liste degisebilir (calisan surum)
List<UnaryOperator<Order>> campaign = List.of(
    OrderEnhancer.giftWrap(),
    OrderEnhancer.insurance(),
    OrderEnhancer.expressShipping()
);
UnaryOperator<Order> enhanceDynamic = campaign.stream()
    .reduce(UnaryOperator.identity(), (pipeline, step) -> pipeline.andThen(step));

var order = new Order("ORD-001", new BigDecimal("200"));
var campaignResult = enhanceDynamic.apply(order);
```

Klasik OOP – Decorator

```java
public interface OrderService {
    Order process(Order order);
}
```

```java
public class BasicOrderService implements OrderService {
    @Override
    public Order process(Order order) {
        return order; // temel islem; ek ozellik yok
    }
}
```

```java
public abstract class OrderServiceDecorator implements OrderService {
    protected final OrderService wrapped;
    protected OrderServiceDecorator(OrderService wrapped) {
        this.wrapped = wrapped;
    }
}
```

```java
public class GiftWrapDecorator extends OrderServiceDecorator {
    public GiftWrapDecorator(OrderService wrapped) { super(wrapped); }
    @Override
    public Order process(Order order) {
        Order processed = wrapped.process(order); // once icteki zincir
        return processed.addFeature("Hediye Paketi", new BigDecimal("15"));
    }
}
```

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OrderService service = new ExpressShippingDecorator(
    new InsuranceDecorator(
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    )
);
```

Hap Bilgi 1

Classic yaklaşımda her yeni özellik için yeni bir wrapper sınıfı gerekir.

Hap Bilgi 2

Modern tarafta `UnaryOperator` zinciriyle davranışlar composable hale gelir.

Modern FP – Decorator

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    return order -> order.addFeature("Hediye Paketi", new BigDecimal("15"));
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    return order -> order.addFeature("Hizli Kargo", new BigDecimal("25"));
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}
```

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public abstract class OrderServiceDecorator implements OrderService {
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    protected OrderServiceDecorator(OrderService wrapped) {
        this.wrapped = wrapped;
    }
}
```

```java
public class GiftWrapDecorator extends OrderServiceDecorator {
    public GiftWrapDecorator(OrderService wrapped) { super(wrapped); }
    @Override
    public Order process(Order order) {
        Order processed = wrapped.process(order); // once icteki zincir
        return processed.addFeature("Hediye Paketi", new BigDecimal("15"));
    }
}
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OrderService service = new ExpressShippingDecorator(
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Hap Bilgi 1

Classic yaklaşımda her yeni özellik için yeni bir wrapper sınıfı gerekir.

Hap Bilgi 2

Modern tarafta `UnaryOperator` zinciriyle davranışlar composable hale gelir.

Hap Bilgi 3

Her iki tarafta da ana nesne bozulmadan özellik eklenir; Open/Closed korunur.

Modern FP – Decorator

```java
// modern/decorator/OrderEnhancer.java
public static UnaryOperator<Order> giftWrap() {
    return order -> order.addFeature("Hediye Paketi", new BigDecimal("15"));
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);
UnaryOperator<Order> enhanceDynamic = campaign.stream()
    .reduce(UnaryOperator.identity(), (pipeline, step) -> pipeline.andThen(step));

var order = new Order("ORD-001", new BigDecimal("200"));
var campaignResult = enhanceDynamic.apply(order);
```

Function Composition Notu

`andThen()` soldan saga uygulanir; zincirdeki sira degistiginde toplam fiyat ve ozellik sırası da degisir.

📋 Template Method Pattern – Rapor Üretimi

Tanım: Template Method, bir algoritmanın iskeletini tanımlar; degisen adimlar alt siniflara ya da fonksiyon parametrelerine birakilir.

Ornek: Rapor uretimi – PDF / Excel / HTML / Email. Iskelet ayni, format degisiyor.

Klasik OOP – Template Method

Modern FP – Template Method

Klasik OOP – Template Method

Modern FP – Template Method

Klasik OOP – Template Method

```
public abstract class AbstractReportGenerator {
    public final void generate(ReportData data) {
        validateData(data);
        String formatted = formatData(data);
        String output = renderOutput(formatted, data.getTitle());
        sendReport(output);
    }

    protected abstract void validateData(ReportData data);
    protected abstract String formatData(ReportData data);
    protected abstract String renderOutput(String formatted, String title);
    protected abstract void sendReport(String output);
}
```

Modern FP – Template Method

Klasik OOP – Template Method

```
public abstract class AbstractReportGenerator {
    public final void generate(ReportData data) {
        validateData(data);
        String formatted = formatData(data);
        String output = renderOutput(formatted, data.getTitle());
        sendReport(output);
    }

    protected abstract void validateData(ReportData data);
    protected abstract String formatData(ReportData data);
    protected abstract String renderOutput(String formatted, String title);
    protected abstract void sendReport(String output);
}
```

```
public class PdfReportGenerator extends AbstractReportGenerator {
    @Override
    protected void validateData(ReportData data) { ... }
    @Override
    protected String formatData(ReportData data) { ... }
    @Override
    protected String renderOutput(String formatted, String title) { ... }
    @Override
    protected void sendReport(String output) { ... }
}
// + ExcelReportGenerator   → 4 method override
// + HtmlReportGenerator    → 4 method override
// + EmailReportGenerator   → 4 method override
// = 16 override, 5 dosya
```

Modern FP – Template Method

Klasik OOP – Template Method

```java
public abstract class AbstractReportGenerator {
    public final void generate(ReportData data) {
        validateData(data);
        String formatted = formatData(data);
        String output = renderOutput(formatted, data.getTitle());
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    protected abstract void validateData(ReportData data);
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    protected abstract void sendReport(String output);
}
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public class PdfReportGenerator extends AbstractReportGenerator {
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    protected void validateData(ReportData data) { ... }
    @Override
    protected String formatData(ReportData data) { ... }
    @Override
    protected String renderOutput(String formatted, String title) { ... }
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    protected void sendReport(String output) { ... }
}
// + ExcelReportGenerator   → 4 method override
// + HtmlReportGenerator    → 4 method override
// + EmailReportGenerator   → 4 method override
// = 16 override, 5 dosya
```

```java
var data = new ReportData("Q1 Satis Raporu", columns, rows);
new PdfReportGenerator().generate(data);
new ExcelReportGenerator().generate(data);
new HtmlReportGenerator().generate(data);
new EmailReportGenerator("yonetim@sirket.com").generate(data);
```

Modern FP – Template Method

Klasik OOP – Template Method

```
public abstract class AbstractReportGenerator {
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```

Modern FP – Template Method

```
public record ReportGenerator(
    Consumer<ReportData> dataValidator,
    Function<ReportData, String> dataFormatter,
    UnaryOperator<String> outputRenderer,
    Consumer<String> reportSender
) {
    public void generate(ReportData data) {
        dataValidator.accept(data);
        String formatted = dataFormatter.apply(data);
        String output = outputRenderer.apply(formatted);
        reportSender.accept(output);
    }
}
```

Klasik OOP – Template Method

```
public abstract class AbstractReportGenerator {
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        String formatted = dataFormatter.apply(data);
        String output = outputRenderer.apply(formatted);
        reportSender.accept(output);
    }
}
```

```
public static ReportGenerator pdf() {
    return new ReportGenerator(
        data → IO.println("  [PDF] Dogrulandi"),
        data → {
            String headers = String.join(" | ", data.columns());
            String rows = formatRows(data, " | ");
            return headers + "\n" + "-".repeat(40) + "\n" + rows;
        },
        formatted → "[PDF_DOCUMENT]\n" + formatted,
        output → IO.println("  [PDF] Yazildi: rapor.pdf")
    );
}
```

Klasik OOP – Template Method

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            return headers + "\n" + "-".repeat(40) + "\n" + rows;
        },
        formatted → "[PDF_DOCUMENT]\n" + formatted,
        output → IO.println("  [PDF] Yazildi: rapor.pdf")
    );
}
```

```
var data = new ReportData("Q1 Satis Raporu", columns, rows);
ReportGenerator.pdf().generate(data);
ReportGenerator.excel().generate(data);
ReportGenerator.html().generate(data);
ReportGenerator.email("yonetim@sirket.com").generate(data);
```

Mesaj

4 format = 4 sinif yerine 4 factory metot. Yeni format? Yeni sinif degil, sadece 4 lambda.

Özet: Ne Öğrendik?

Özet: Ne Öğrendik?

Builder

Verbose builder → Immutability & records

Özet: Ne Öğrendik?

Builder

Verbose builder → Immutability & records

Strategy

Interface + class → Lambda ifadesi

Özet: Ne Öğrendik?

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Verbose builder → Immutability & records

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Interface + class → Lambda ifadesi

Decorator

Class hiyerarşisi → andThen() kompozisyonu

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Abstract class → Higher-order functions

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Builder

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Interface + class → Lambda ifadesi

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Template Method

Abstract class → Higher-order functions

Modern Java'nın Kazanımları

- ✓ %60-80 daha az kod
- ✓ Daha yüksek okunabilirlik
- ✓ Daha kolay test edilebilirlik
- ✓ Yüksek composability – küçük davranış parçalarını (lambda, `andThen`, HOF) birleştirerek yeni akışlar kurmak

Özet: Ne Öğrendik?

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Verbose builder → Immutability & records

Strategy

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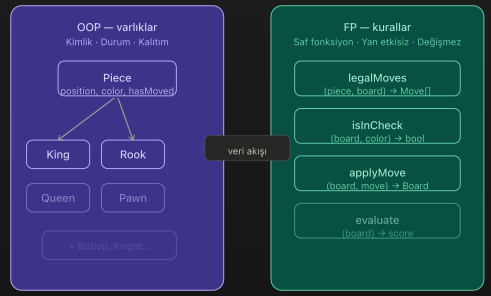
Kullanılan Java Özellikleri

- Lambda Expressions
- Stream API
- Records
- Pattern Matching
- Functional Interfaces
- Method References
- Sealed Classes

OOP vs FP – karar çerçevesi

AYNI DOMAIN – IKI ROL



OOP → kimlik · durum FP → saf fonksiyon

OOP "neyi" · **FP** "nasıl"

Doğru aracı doğru katmana vermek.

♝ OOP – kimlik ve durum

- Taşlar *nesnedir*; konum, geçmiş, "rok oldu mu?" gibi **durum** vardır.
- `King` , `Rook` , `Pawn` … `Piece` özelleşmesi – kalıtım burada doğal.
- OOP: **kimlik ve durum**.

λ FP – kurallar ve dönüşüm

- Hamle, şah, mat: **evrensel** kurallar (kişisel tercih değil).

> Don't be a functional programmer.
>
> Don't be an object-oriented programmer.
>
> **Be a better programmer.**

Brian Goetz

FP vs OO: Choose Two

> *Any fool can write code that a computer can understand. Good programmers write code that humans can understand.*

Martin Fowler

Refactoring: Improving the Design of Existing Code

Teşekkürler!

Modern Java ile Design Patterns'ın Fonksiyonel Evrimi

GitHub

`github.com/umiitkose`

GitHub QR

YouTube

`Design Patterns Serisi`

YouTube QR

Örnek Kodlar

`javadayistanbul-functional-evolution-of-design-patterns-with-modern-Java`

Repo QR

Sorularınız?